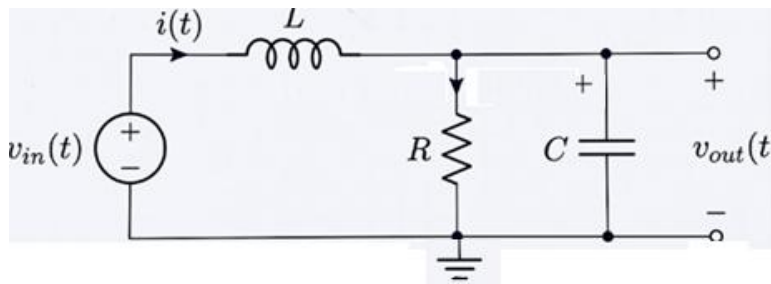


A second -order electrical circuit is shown in the figure below, where v_{in} is the input and v_{out} is the output. $L = 1$ H, $C = 1$ F. Do the problem for $R = 1 \Omega$ and $R = 0.25 \Omega$



1. Find the network function of the circuit.
2. Determine the poles of the network function.
3. Find the state variable representation of the circuit. Take $i(t)$ as x_1 (with the current direction shown in the figure) and the voltage across the capacitance as x_2 (with the polarity shown).
4. Determine the eigenvalues of the A matrix
5. Obtain the network function from the state variable representation.
6. What is the natural frequency and damping ratio when $R = 1 \Omega$ and $R = 0.25 \Omega$
7. Using the network function approach, find the unit step response of the circuit. $R = 1 \Omega$
8. Using the state variable representation, assuming the initial conditions are zero, determine unit step response of the circuit.
9. Find the equilibrium point from the state equations when $R = 1$
10. With the help of the figure, justify the answer obtained in question 9.

Extra question (No marks): What is the difference in the output (unit step response) when $R = 1 \Omega$ and $R = 0.25 \Omega$

Notes:

Question 1 can be solved easily. Normally we write the equations. Then take Laplace transform. Then find the ratio of output transform to input transform. But here it is a simple circuit.

$$V_o(s) = Z_2(s) I_1(s)$$

$$V_{in}(s) = (Z_1(s) + Z_2(s)) I_1(s)$$

$$N(s) = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$$

Here $Z_1 = Ls$
 $Z_2 = R \parallel \frac{1}{Cs}$

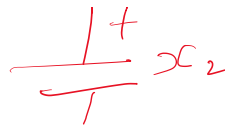
Poles are roots of the denominator polynomial (Question 2). In question 3, you have to write equations in the following form:

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $u = \text{input}$
 $y = o/p$

Since there are 2 (n) energy storage elements, the order of A matrix is 2x2. Only one input (r). The order of B matrix is 2x1 (nrx). There is only one output (m). Hence the order of C matrix is 1x2 (mxn). C is row matrix. Similarly the order of D is 1x1 (mxr).



The first step is to select the state variable. Select x_1 as inductor current and x_2 as capacitor voltage. Important step is to assign a direction for inductor current and a polarity for the capacitor voltage. Then write KVL in the inductor loop and write KCL at the capacitor junction. We do this to get the state equations in the form given in the last page directly.

$$\dot{X} = AX + Bu \quad \dot{x}_1 = \dots$$

$$\dot{x}_2 = \dots$$

Write these two first order equations in matrix form, So you got A and B matrices

Now write the output y in terms of state variables and get the output equation in matrix form. $Y = CX + Du$

Poles of network function are the same as eigenvalues of A matrix. But you have to revise eigenvalues and eigenvectors.

You can find the transfer function (question 5) using the formula in slide 21 of lecture 8- slides

$$N(s) = C(sI - A)^{-1}B + D$$

Equate the network function to general form of second order transfer functions.

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Then find the natural frequency and damping ratio.

Question 7 and 8 are the main questions. The unit step response is the output y when the input is a unit step voltage. You have to get the answer by the same method



$$V_i(s) = \frac{1}{s}$$

$$V_o(s) = N(s) V_i(s)$$

$$V_o(t) = \mathcal{L}^{-1} \left(N(s) V_i(s) \right)$$

$$\dot{X} = AX + Bu$$

State eqn

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

When you solve state equation, you get the state X (set of state variables).

Substitute the state variables in the output equation. $Y = CX + Du$

To solve the state equation, you can use the formula (slide 20 of lecture 8)

$$u(s) = \frac{1}{s} \quad X(t) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathcal{L}^{-1} \left((sI - A)^{-1} X_0 + \mathcal{L}^{-1} (sI - A)^{-1} B u(s) \right)$$

In the formula, the first part is the response due to initial conditions (here this part is zero because the initial conditions are zero). The second part is the response due to input.

To determine the equilibrium points, equate the LHS of state equations to zero. For what values of x_1 and x_2 , you will be getting the derivatives of state variables equal to zero. You can verify this by knowing the behaviour of C and L when time tends to infinity.

$$\dot{x}_1 = 0$$

$$\dot{x}_2 = 0$$